

Solar Fleet Intelligence

Fleet Performance Report — TPREL

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Model: XGBoost | $R^2 = 0.9004$ | MAE = 0.0083
Inverters monitored: 10
Fleet annual loss: ■555,289

1. Executive Summary

Out of 10 inverters analyzed, **3** show elevated anomaly rates above the fleet baseline. These inverters account for the majority of the ■555,289 annualized fleet loss. Immediate maintenance attention is recommended for the top 3 underperformers.

2. Fleet Health Overview

Inverter	Anomaly Rate (%)	Avg Efficiency	Avg Residual
INV-009	3.54	0.8969	-0.002992
INV-007	2.93	0.9003	-0.002710
INV-003	2.15	0.8471	-0.002312
INV-005	1.98	0.9271	0.001181
INV-002	1.77	0.9272	0.001154
INV-008	1.73	0.9278	0.001370
INV-001	1.60	0.9270	0.001148
INV-010	1.58	0.9277	0.001275
INV-006	1.43	0.9270	0.001040
INV-004	1.29	0.9271	0.001108

3. Cost Impact Analysis

Assumptions: Tariff = ₹8.0/kWh, Nominal Efficiency = 95%. Values represent annualized revenue loss per inverter.

Inverter	Avg Eff.	Energy Loss (kWh)	Annual Loss (₹)
INV-003	0.8471	19,366	₹154,927
INV-009	0.8969	10,153	₹81,222
INV-007	0.9003	9,649	₹77,188
INV-002	0.9272	4,361	₹34,891
INV-008	0.9278	4,344	₹34,753
INV-010	0.9277	4,340	₹34,719
INV-001	0.9270	4,322	₹34,575
INV-004	0.9271	4,315	₹34,519
INV-005	0.9271	4,282	₹34,257
INV-006	0.9270	4,280	₹34,238
TOTAL	—	—	₹555,289

4. Recommendations

- **INV-003** — Likely soiling issue. Schedule panel cleaning within 1 week.
- **INV-009** — Possible sensor fault. Verify sensor calibration.
- **INV-007** — Aging-related degradation. Consider component inspection.
- Monitor fleet baseline weekly; any inverter exceeding 2% anomaly rate should be escalated.
- Invest in cooling/ventilation if temperature-driven losses persist.

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